

Data Professional Survey Dashboard

Project Overview

Developed an interactive Power BI dashboard to analyze survey responses from 630+ data professionals across different countries, industries, and job roles. The project involved cleaning and transforming raw survey data using Power Query and creating visualizations to uncover trends in salaries, career transitions, job satisfaction, and technology preferences.

Github link for the project can be found here:

<https://github.com/aishwaryaSingh2804/Data-Professional-Survey-Dashboard/tree/main>

Tools Used: Power BI, Power Query, Microsoft Excel

Data Cleaning and Transformation

To improve data quality and reporting accuracy, the following transformations were performed:

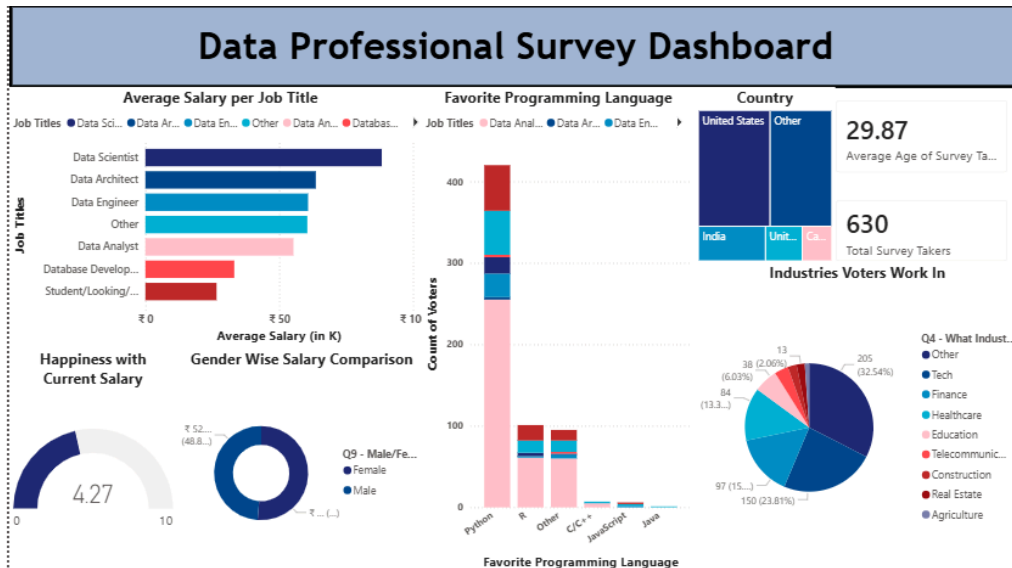
- Removed irrelevant and empty columns such as Email, Browser, Operating System, City, Country metadata field, and Referrer.
- Standardized inconsistent categorical values across multiple fields.
- Consolidated "Other" responses into meaningful categories for better analysis.
 - Example: "Other (Please Specify): Retail" → "Retail"
 - Example: "Other: Sql" → "SQL"
- Cleaned text fields by removing inconsistencies and duplicate category variations.
- Validated data types for numerical, date, and categorical fields.
- Reviewed records for duplicates and missing values to ensure data integrity.

Dashboard Highlights

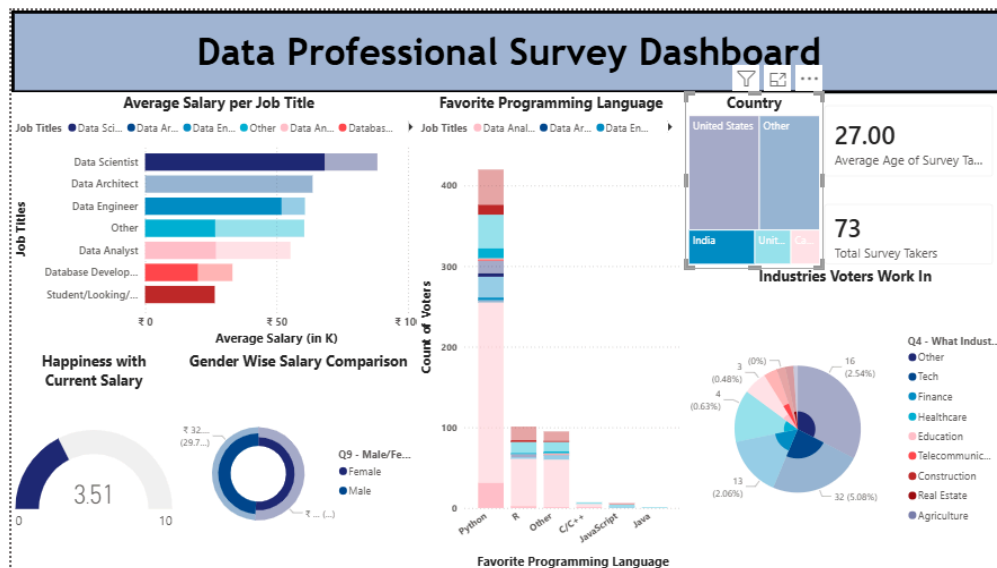
The dashboard was designed to provide a comprehensive overview of survey respondents through:

- Salary Distribution Analysis
- Favorite Programming Language Analysis
- Career Transition Analysis
- Job Satisfaction Analysis
- Demographic Analysis (Age, Gender, Education, Ethnicity, Country)

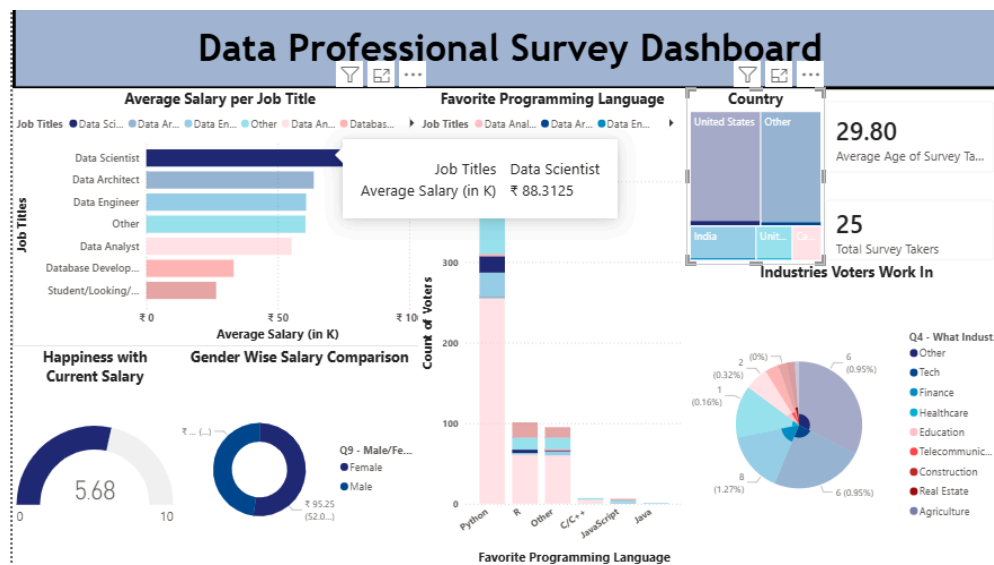
Interactive filters and cross-filtering capabilities allow users to explore insights dynamically across different respondent groups.



Dashboard showing different KPI insights



Cross filtering done on the basis of county: India



Cross Filtering done to show KPI insights for data scientists

Key Insights

- Python emerged as the most preferred programming language among data professionals.
- A significant proportion of respondents transitioned into data careers from other fields.
- Salary satisfaction scored lower compared to work-life balance and coworker satisfaction, indicating compensation as a major concern.
- Data Analysts and Data Scientists represented the largest respondent groups.
- Salary distributions varied considerably across job roles and countries.
- Better salary, remote work opportunities, and career growth were among the most important factors when considering a new job.
- Survey responses highlighted differences in satisfaction levels across demographic groups and professional backgrounds.

Outcome

The project demonstrates practical experience in data cleaning, transformation, dashboard development, and business insight generation using Power BI. The dashboard converts raw survey data into actionable insights that support data-driven decision-making.